

How Should We Allocate the Marketing Budget?

A Bayesian Media Mix Modeling Approach to Revenue Measurement and Budget Optimization

13 weeks
budget horizon

+10.80%
expected revenue

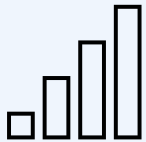
8 mixed
media channels

\$5.18 million
media budget



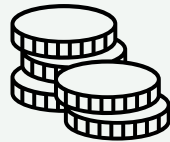
How should we allocate the next marketing budget?

We have a fixed 13-week marketing budget across eight media channels. The goal is to find the allocation that maximizes expected revenue.



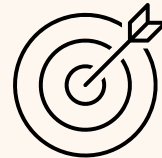
What is driving revenue today?

Which channels are contributing most to revenue?



Where should the next dollar go?

Which channel currently has the highest marginal return?

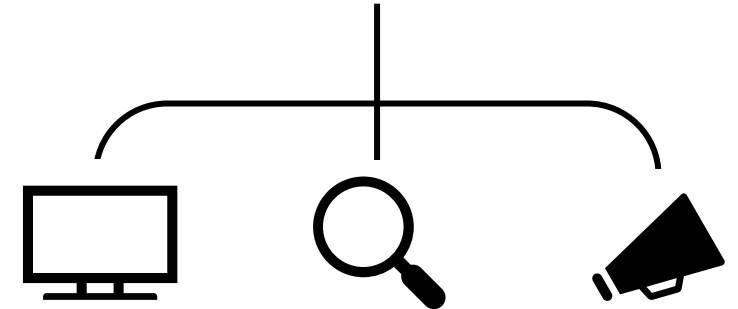


How should we allocate the full budget?

What mix is expected to generate the highest revenue?

Total Marketing Budget

\$5.18M



Video/TV
Television, Video-on-demand

Digital
Google, Display, Facebook, Amazon

Offline
Out-of-home, Radio



Maximize Revenue

Why is budget allocation difficult?

More channels and more money do not automatically translate into higher revenue.



John Wanamaker (1838-1922)
pioneer in marketing

Half the money I spend on advertising is wasted; the trouble is I do not know which half.

Media effects are nonlinear

Advertising can have delayed carryover and diminishing returns as spend increases.

Historical data is limited

Newer channels, such as TikTok, have shorter histories and fewer observations to learn from.

Attribution is messy

Promotions, seasonality, trends, and multiple channels can move revenue at the same time.

Bayesian Marketing Mix Modeling (MMM)

Business Data



Media Spend

Digital: Google Search, Display and Video 360, Facebook, Amazon Marketing Services

Video/TV: Television, Video-on-demand

Traditional Offline: Out-of-home, radio

Other Drivers

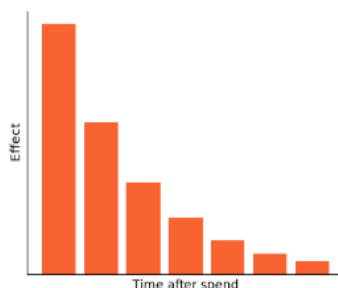
- Promotions
- Seasonality
- Trend



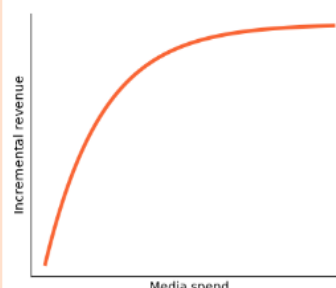
BAYESIAN MMM

Separates media impact from other business drivers

Carryover



Saturation



Used in practice by companies such as HelloFresh, Bolt, and Walmart

Business Decisions

01

What drove revenue?

Channel contribution and historical ROAS

02

Where should the next dollar go?

Marginal ROAS and response curves

03

How should we allocate the budget?

Optimized media mix

WHAT HAPPENED?

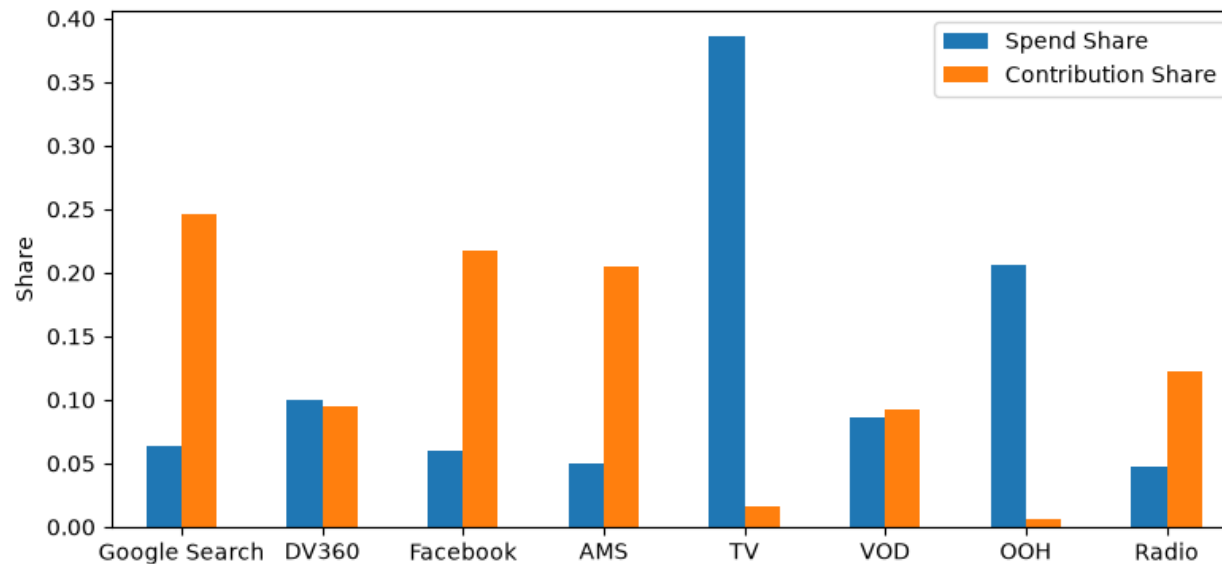
WHY DID IT HAPPEN?

WHAT SHOULD WE DO NEXT?

What is driving revenue today?

Historical spend, modeled media contribution, and ROAS point to a clear mismatch in the current mix.

Spend share vs. modeled media contribution



AMS + Search + Facebook
≈18% of spent → ≈67% of contribution

TV + OOH
≈59% of spent → ≈2% of contribution

Historical ROAS

Modeled revenue per \$1 of historical media spend

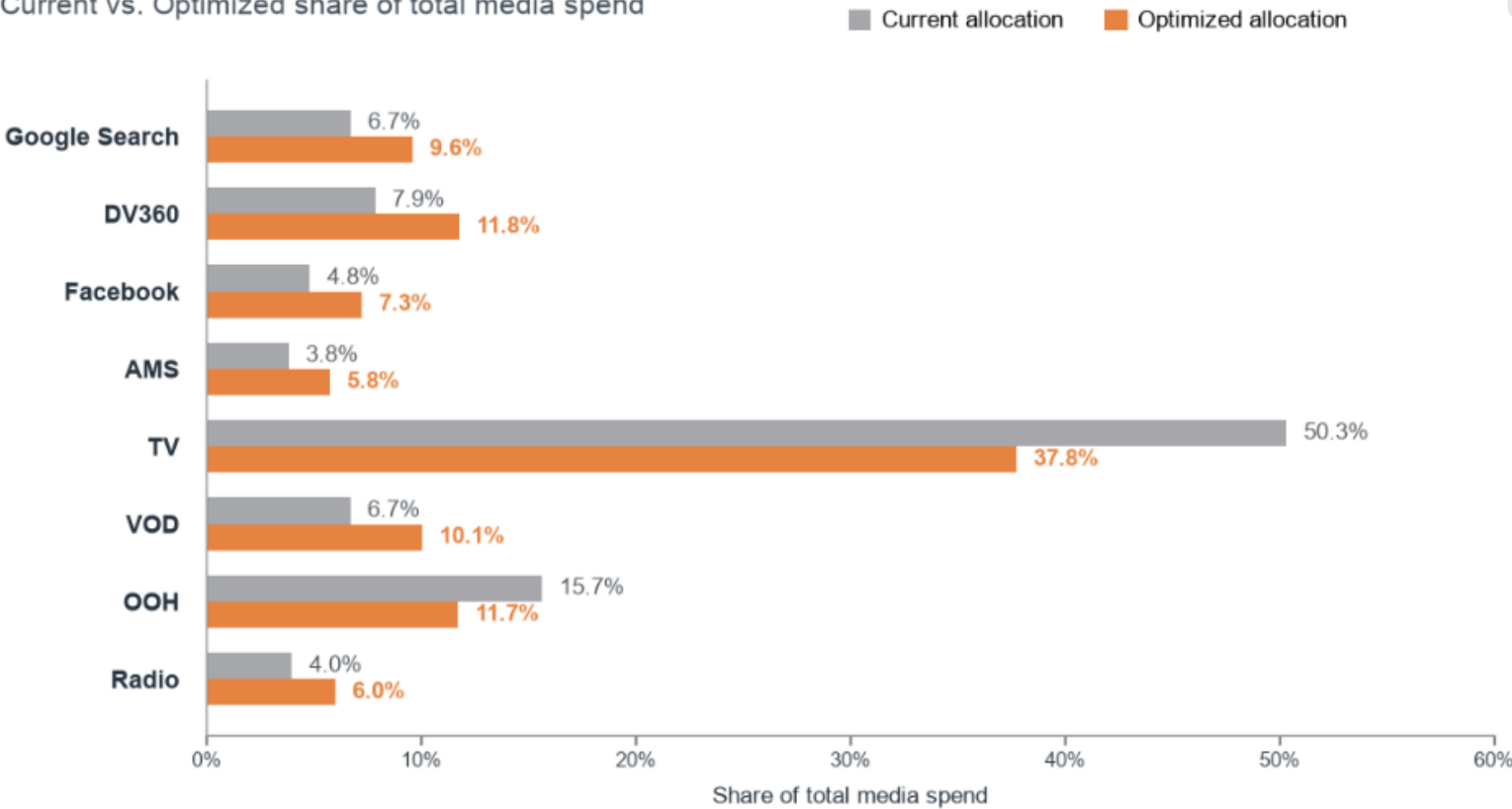
Channel	ROAS
AMS	22.91x
Google Search	21.74x
Facebook	20.24x
Radio	14.49x
VOD	6.07x
DV360	5.31x
TV	0.21x
OOH	0.13x

Same budget, better mix.

Reallocate the next 13 weeks within practical channel guardrails.

Channel Allocation

Current vs. Optimized share of total media spend



Fixed 13-week media budget | **\$5.18M**

Practical channel change guardrail | **±25%**

Objective | **Maximized expected revenue**

Expected Revenue

\$10.76M → \$11.92M

+10.80%
expected lift

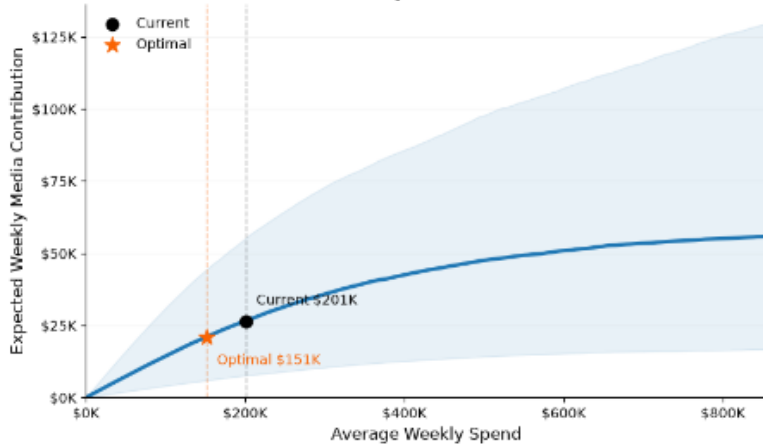
Key Takeaway

Shift approximately **\$850K** out of TV and OOH and redistribute it across the other six channels – without increasing total spend.

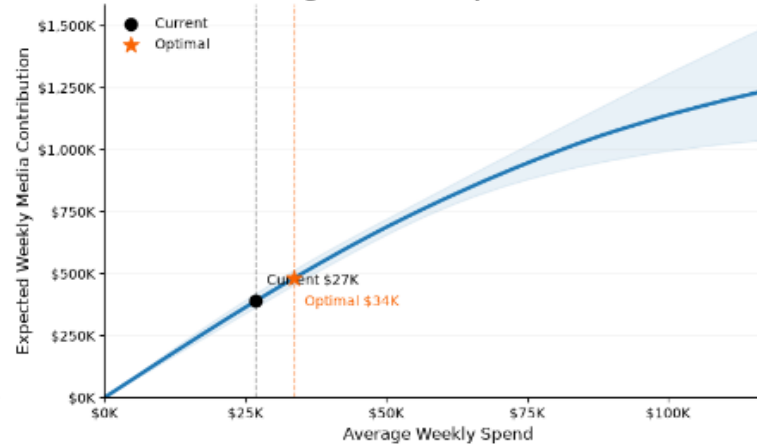
Why this allocation?

The channels are at different points on their response curves.

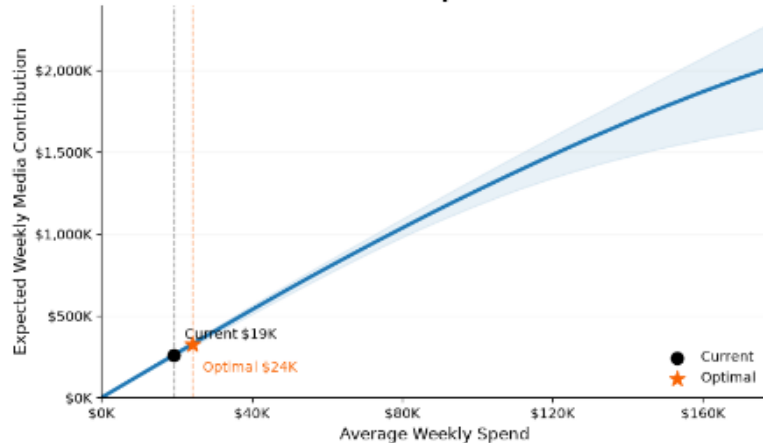
TV Response Curve



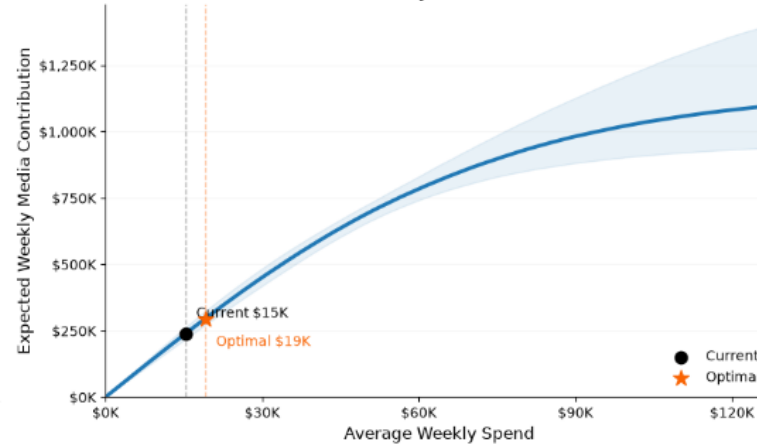
Google Search Response Curve



Facebook Response Curve



AMS Response Curve



What the response curves tell us

TV has less room at the margin

At the current spend level, the TV response curve is already relatively flat. Reducing spend modestly gives up comparatively little expected contribution.

Search, Facebook, and AMS still have headroom

Their current spend levels sit on steeper parts of the response curves, where additional budget can still generate meaningful incremental contribution.

Why the mix shifts

Move some budget from TV and OOH to channels where the next dollar is expected to work harder

The recommendation stays within the $\pm 25\%$ channel guardrails.

From Model Recommendation to Action

\$850K model shift out
of TV and OOH



Model Target

Keep total budget fixed at \$5.18M

TV 50.3% → 37.8% ≈ \$643K
OOH 15.7% → 11.7% ≈ \$207K

Shifted to other channels ±25%
guardrails

Validation Pilot

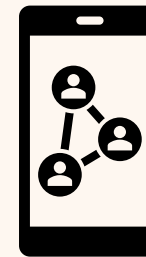
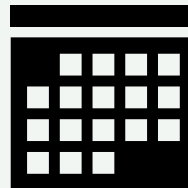
Weeks 1–4 Test 20% of the
recommended shift in selected
markets

Weeks 5–8 Increase to 50% if
incremental revenue and ROAS
improve

Weeks 9–13 Scale toward the full
optimized mix if the lift persists

Geo Holdout

Staged Rollout



New Social Media

Run a randomized geo test for
TikTok

Measure incremental lift

Add to the MMM once validated

Thank you for your attention!

Phase 1 Data Preparation



Data setup



365 weekly observations



8 media channels



Controls: promotion, trends, seasonality



Train/holdout split: 313/52 weeks

Prepared



Phase 2 Model Fitting



Bayesian MMM



PyMC-Marketing implementation



Geometric Adstock + logistic saturation



MCMC sampling completed



Max R-hat = 1.004

Fit Complete



Phase 3 Model Diagnostics



Check Model Behavior



Posterior checks and response curves



Channel contribution estimation



ROAS and mROAS



Uncertainty intervals reported

Checked



Phase 4 Attributions & Optimization



Optimize decisions



Fixed 13 week budget: \$5.18M



Channel change bounds: 25%



Optimized relocation across channels



Model revenue uplift: 10.80%

Optimized



Phase 5 Delivery & Next Steps



Decision ready output



Github workflow



Next steps: staged rollout & validation



New social media: Tiktok



Deployment after broader validation

Ready for next step